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Beyond topography: addressing the functional impact of social skills training for students with autism

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ABSTRACT

Students with autism spectrum disorder often require social supports to function well in a mainstream class setting. Researchers have demonstrated the effects of social skills training for students with autism, but their work has focused primarily on topographically defined outcomes, such as increasing the frequency of social behaviours. More research is needed to address the functional impact of social skills training on social experiences, such as improvements in the quality of life for students with autism. This study examined the effects of a behavioural skills training form of social skills intervention on the social interactions of students with autism in an inclusive school setting. The intervention increased participants' social behaviour. However, its mixed results in the quality of peer responses may be a more meaningful indicator of its effect on the quality of social lives of the participants. The authors discuss the implications of these findings for current practice and future research.

KEYWORDS

Autism; social skills training; primary school; functional intervention

Introduction

From its inception in the 1980s, the positive behaviour support (PBS) movement has been dedicated to improving the quality of life of individuals, especially those with disabilities (Kincaid et al., 2015). PBS is a branch of behavioural psychology, with roots in the philosophy of radical behaviourism and the principles and practices of applied behaviour analysis (Singer & Wang, 2009). PBS places paramount importance on maintaining high moral ideals such as respecting personal dignity, encouraging autonomy, and promoting respect. In a recent effort to define PBS, leaders in the field referred to it as an approach to improving behavioural outcomes for individuals with and without disabilities by focusing on prevention, reinforcement, quality of life, and individual dignity (Kincaid et al., 2015). This approach has been popularised in the US and internationally by the Positive Behavioural Interventions & Supports Technical Assistance Centre (pbis.org).

Individuals with Autism Spectrum Disorder (ASD) have recently received increased attention, both within and beyond the PBS community. As social interaction is a prevalent issue for those on the spectrum, a number of social skills interventions have been developed to try to improve the social aspects of their lives (for example, Laugeson, Gantman, Kapp, Orenski, & Ellingsen, 2015). Since 2014 at least six systematic reviews or

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meta-analyses have been published endorsing social skills interventions specifically for children with ASD (Camargo et al., 2014; Goldstein, Lackey, & Schneider, 2014; McCoy, Holloway, Healy, Rispoli, & Neely, 2016; Watkins et al., 2014; Whalon, Conroy, Martinez, & Werch, 2015 ; Wong et al., 2015). From their review, Camargo et al. (2014) concluded that there was sufficient evidence for behaviourally-based social skills interventions to be considered an evidence-based practice. Goldstein et al. (2014) also reviewed social skills interventions for preschool students and found substantial evidence to support these early interventions.

Although increasing evidence supports the use of social skills training to increase social performance, little evidence has demonstrated that this improved performance actually improves the quality of the social lives of individuals with ASD. This discrepancy is at least partially the result of how the dependent variables have been defined. Researchers typically assess the effectiveness of an intervention by measuring changes in terms of topography; what the behaviour looks like. Examples of these dependent variables include social initiations, social responses, conversations, and peer imitation (Camargo et al., 2014; McCoy et al., 2016). Although improvements in behavioural topographies (for example, joining a game, greeting a peer) may be a good outcome, such improvements may not equate to a functional improvement in the social life of the child with ASD: positive change in the student's environment (for example, more reinforcing relationships, better acceptance among peers).

From a behaviour analytic perspective, there are three possible effects that a behaviour can produce; (a) *reinforcement*, thereby strengthening that behaviour in the future under similar circumstances, (b) *punishment*, thereby weakening that behaviour in the future under similar circumstances, (c) or no change, referred to as *extinction*, thereby resulting in a more gradual weakening of that behaviour in the future under similar circumstances. Behaviours that result in reinforcement may be said to produce a functional improvement. Use of the term *function* in the current study is broader than, but related to, how the word is used in the context of a functional behaviour assessment or functional analysis. In that context *function* simply means the reinforcer that maintains a specific behaviour. For the purposes of this paper, social skills will be discussed in terms of all of the possible effects of those behaviours (that is, reinforcement, punishment, and extinction).

Elliott, Racine, and Busse (1995) defined social skills as "socially acceptable learned behavior that enable a person to interact with others in ways that elicit positive responses and assist in avoiding negative responses" (p.1009). The "socially acceptable learned behavior" aspect of this definition may be captured by a topographical description of a behaviour but miss the functional, quality of life, portion: "elicit positive responses and assist in avoiding negative responses." Neglecting the functional aspect of the definition is no trivial oversight, as the same topography of behaviour may result in either positive or negative social interactions. For example, a child with ASD may learn the skill of joining a game and perform the skill accurately, resulting in acceptance and appropriate play. However, using that same skill might alternatively result in the child being excluded from the game or accepted only to provide a target for teasing or bullying. In both cases, an observer could indicate that the child with ASD demonstrated an improvement in social skills (that is, increased frequency), but it does not necessarily follow that in both cases the quality of the child's social life has been improved.

To confirm the effectiveness of a social skills intervention for improving social lives of children with ASD, researchers need to measure and report both its topographical and functional effects. The functional effects of a social skills intervention might be measured in terms of how typically developing peers respond when students with ASD perform newly acquired social behaviours. For example, after teaching a student how to join a game, researchers could measure topographical improvements in terms of frequency and accuracy of using the new skill and measure functional improvements by assessing peers' positive responses (for example, accepting and interacting appropriately) or negative response (for example, rejecting, mocking, or bullying). The dual methodology would better address the Elliott et al. (1995) definition of social skills and a fundamental goal of PBS: to improve the quality of the lives of indivdiuals with disabilities.

The purpose of the current study was to address both the topographical and functional effects of a social skills training intervention. To do this the authors implemented a brief behavioural skills training (BST) intervention to improve social skills with primary school students with ASD (Miltenberger, 2015). Three research questions were addressed:

To what extent does a BST social skills training approach increase the frequency of appropriate social vocalisations (topographically defined) for primary students with ASD?

To what extend does a BST social skills training approach increase the occurrence of positive and negative responses from typically developing peers (functionally defined)?

To what extent are the topographical outcomes (social vocalisations) consistent with the functional outcomes (peer responses)?

Method

Participants

Three primary school students (all names are pseudonyms) with Autism Spectrum Disorder (ASD) participated in this study (see Table 1). All met the following inclusion criteria: (a) had a special education classification of ASD, (b) were in Grades 2–5 (ages 7–11), (c) participated in the mainstream class at least 50% of the school day, (d) engaged in at least 25% fewer appropriate social vocalisations than typically developing peers (see below), (e) scored below the 25th percentile on the Social Skills Improvement System (SSIS) social skills rating scale, and (f) had verbal communication skills approximately commensurate with their same-grade peers as indicated by teacher perception.

Prior to recruiting participants, this study was approved by a university Institutional Review Board and approved by the school district's research review process. Additionally, assent was obtained from each student and written informed consent was obtained from their respective parents. Finally, the authors carefully monitored the impact of the intervention to determine if it was having any adverse effects on the participants (for example, increased incidents of bullying).

Table 1. Participant information.

Participant	Sex	Grade	Race/Ethnicity	SPED Services Received	SSIS Social Skills Scale Percentile Rank
Juan	Male	3	Latino	Reading, Maths	11
Doug	Male	3	Caucasian	Reading	8
Elli	Female	5	Caucasian	Speech, Reading	5

SPED = Special education, SSIS = Social Skills Improvement System. All names are pseudonyms.

Juan spent a good portion of his time (not exceeding 50%) in the resource room with other students receiving services for reading and maths, some of whom were also in his mainstream class. Thus, much of his time was spent with other students who were receiving special education services. Doug spent more than 75% of his time in the mainstream class and received reading support in the same reading group as Juan in the resource room. Elli spent nearly all her time in the mainstream class but received some services in a withdrawal from class setting for speech and reading.

Interventionists

This study included two interventionists (non-researchers), both Caucasian females, who implemented the BST social skills training. One was a speech pathologist with a master's degree and 18 years of experience working in schools; the other was a first year special education teacher, with a bachelor's degree. Both of the interventionists had completed course work in serving students with disabilities; however, neither had any focused training on working with students with ASD. They were both selected because they were available at the time of day the intervention needed to occur. Both received training on how to implement the BST social skills lessons and demonstrated competence prior to delivering the intervention to students.

Setting

This study was conducted in two primary schools (K-5) across two rural school districts in the USA. The intervention took place in two settings in each school: (a) a classroom, office, or hallway without distractions for delivering the intervention and (b) the school playground, the target environment for the social skills training. The researcher estimated that as many as 100 students across three grades were typically on the playground during lunch recess, with between two and five playground supervisors and occasionally one or two parent volunteers. A data collector was also present. Data on the dependent variables were collected on the playground. The cafeteria, which was similar to the playground with respect to the composition of students and adults, served as a generalisation setting.

Data collection

Direct observation data were collected on smartphones using the Behavioural Observation Tool (BOT, www.simontonsoftware.com/BOT). The BOT system time stamped each data point so that it could be temporally compared to other data points to calculate interobserver agreement (IOA). Each observation lasted 10 minutes. Additionally, peer composite data were collected on typically developing peers to provide a socially valid comparison. These data were collected by selecting from a convenience sample of same gender, same grade peers on the playground and collecting direct observation data for two minutes, then moving to another peer and continuing this process until the 10-minute observation was completed. Using the peer composite data, a 95% confidence interval was calculated for the peers of each participant, as represented by a grey band on each graph (see Figure 1).

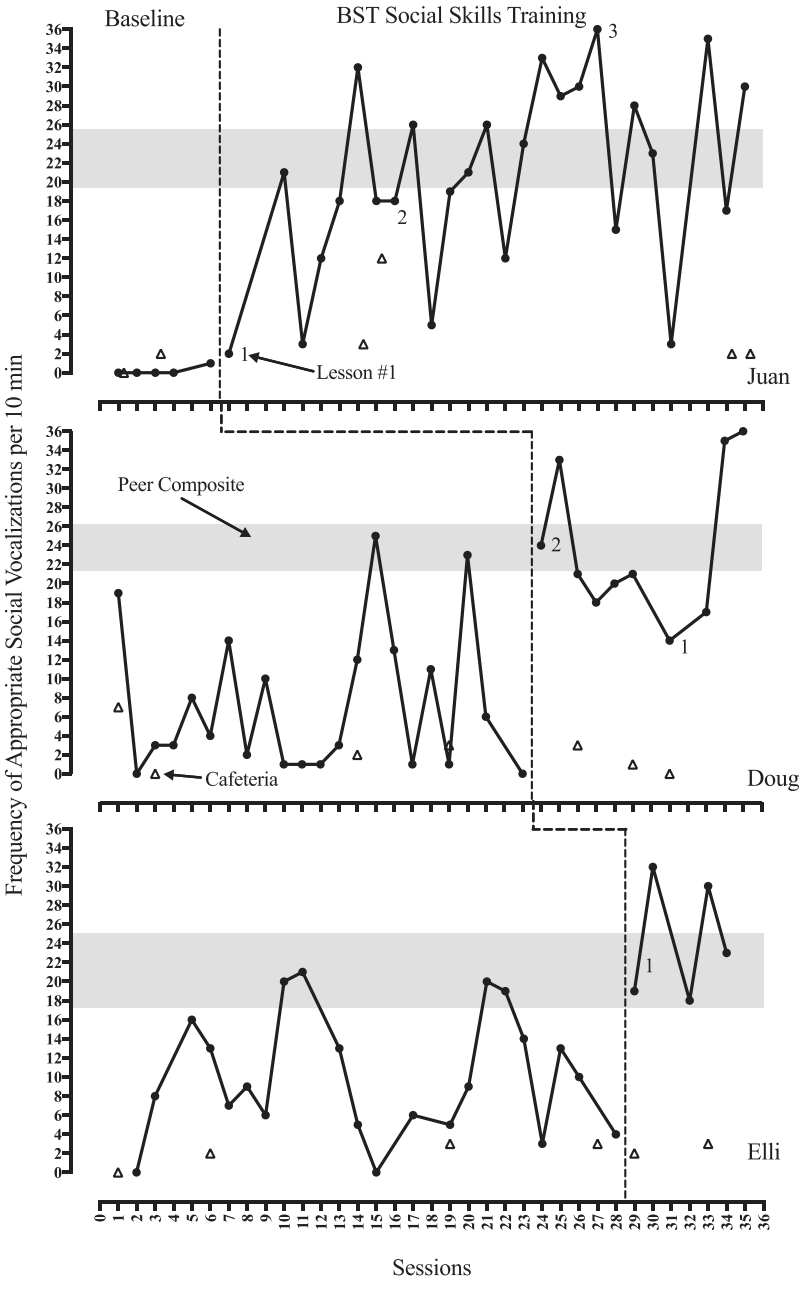


Figure 1. Frequency of appropriate social vocalisations.

Dependent variables

Appropriate social vocalisations

Appropriate social vocalisations were defined as physically orienting towards typically developing peer(s) (for example, face, eyes, or torso pointed towards peer) and emitting a positive vocalisation (for example, calling the person's name or saying something like "hi," or "pass me

the ball") or inviting the peer to participate in an activity, a conversation, or a reciprocal salutation. Non vocal behaviours and vocal behaviours discouraging positive peer responding, (for example, name calling or making comments like "give me that ball back," "get out of the way," "stop doing that") were not counted as appropriate social vocalisations. Social vocalisations were measured using a frequency count during each 10-minute observation.

Peer responses

Peer responses to social vocalisations included positive peer responses, negative peer responses, and no peer response. *Positive peer responses* were defined as positive vocal or non vocal behaviours in response to a social interaction initiated by a participant (for example, saying "sure, let's play," passing the participant the ball, looking at the object indicated by the participant). *Negative peer responses* were recorded for any vocal or non vocal response that was physically aggressive (for example, hitting, kicking, pushing), verbally aggressive (for example, "shut up," "you're an idiot,"), or relationally aggressive (for example, "no way," "go away," running away), including active refusal to engage with the participant (for example, "no thanks," "not today," "I don't want to play that game"). N*o peer response* was defined as the absence of a positive or negative peer response. For this passive category, data collectors did not record anything. *No peer response* was recorded as a passive category because the pilot studies suggested that data collectors struggled to accurately collect data on all the variables of interest. Having a passive category reduced the burden. *Positive* and *negative* peer responses were measured using a partial-interval recording system. The interval began only when a participant began a social vocalisation, and the interval ended five seconds after the vocalisation ended. A *positive* or *negative* peer response occurring in the time between the participant's initial vocalisation and 5 seconds after its termination was recorded.

Independent variable

The social skills training implemented a behavioural skills training process (Miltenberger, 2015) with some additional features designed to improve on previous social skills training efforts. The process included assessment, lesson planning, lesson delivery and reinforcement/progress monitoring. Woven into these steps were (a) an effective behavioural skills teaching cycle (that is, direct instruction, teacher modelling, student rehearsal, and teacher feedback), (b) daily lessons lasting approximately 15 min, (c) instruction occurring in the setting relevant to the skill (for example, play skills on playground), (d) individualised lessons, and (e) progress monitoring.

Preliminary assessment

The assessment prior to this intervention consisted of two components: (a) direct observations to determine that the student produced at least 25% fewer social behaviours than typically developing peers, and (b) teacher completion of the Social Skills Improvement System (SSIS) Teacher Form (Gresham, Elliott, & Kettler, 2010). The SSIS measured strengths and deficits to guide lesson planning. Based on teacher responses on the SSIS, the interventionists and researcher developed lesson plans around three skills: (a) starting a conversation (indicated by numeral 1 in Figure 1), (b) joining a game that is in progress (numeral 2), and (c) asking others to join a game (numeral 3). These skills were selected because they were deficits shared by all participants and closely related to the primary dependent variable.

Lesson planning

Once specific skills were identified and prioritised, the interventionists and researcher created a lesson plan for each skill to be learned. Lesson plans included a brief introduction, which included a check for prerequisite skills and a follow-up on previous goals as relevant. After the introduction, the lesson followed a behaviour skills training process that included direct instruction, teacher modelling, student rehearsal, and teacher feedback. The final step was setting a goal for the participant to engage in independent practice of the skill during recess.

Lesson delivery

To deliver the lesson, the interventionist excused the participant from the cafeteria approximately five minutes before the beginning of lunch recess and took the child to a classroom or office to deliver the introduction, instruction, modelling, and initial rehearsal with feedback (for a total of two – seven minutes). The participant and interventionist then moved to the playground, the natural context for the new skill, and continued the rehearsal and feedback. The participant rehearsed the skill with the interventionist and a variety of typically developing students until mastery was demonstrated (three demonstrations at 100% accuracy; three – nine minutes).

After setting a goal for a minimum number of times the participant was to use the skill during the remainder of recess, but after encouraging the student to use it more as required, the interventionist left the student to apply it independently (time 1–2 minutes). In total, the lesson delivery took no more than 15 minutes per day. Lessons were delivered daily until the student had mastered all skills on the prioritised list.

Reinforcement and progress monitoring

As students with ASD are often not motivated to engage in social behaviour, even with skills they can successfully perform, reinforcement and progress-monitoring components were included in the intervention. The data collector observed the participant during lunch recess and recorded the number of appropriate social vocalisations to report to the interventionist. The interventionist graphed the results and showed them to the participant during the following lesson. Participants who met their goal were given access to items or activities from a reinforcement menu and were required to invite a typically developing peer to join them in accessing the reinforcer.

Experimental design

A concurrent multiple baseline design across participants (Cooper, Heron, & Heward, 2007) was used to evaluate the effects of BST social skills training on participants' social behaviour. A multiple-baseline design was selected because the intervention focused on skill acquisition, which is not likely to be reversible.

Procedures

Baseline phase

The researcher conducted preliminary observations of potential participants to ensure that they met the inclusion criteria of engaging in 25% fewer appropriate social vocalisations than their typically developing peers and scoring below the 25th percentile on the

SSIS. Data collectors observed each of the three participants on the playground daily for 10 minutes during lunch recess. Also data collectors recorded data on peers as a composite during lunch recess. They chose one student who was the same gender and approximately the same age as the participant and observed that student for two minutes, then selected another student and repeated the process to observe five students. During this phase, study participants were involved with only those social skills interventions provided class-wide.

Intervention phase

After the lesson plans were fully developed, the interventionists administered the first few sessions with the researcher so that the researcher could model the intervention and gradually turn it over to the interventionist while providing feedback. When the interventionist reached 100% fidelity, the researcher no longer participated unless scheduling conflicts required him to substitute. Neither of the interventionists required more than five sessions with the researcher to achieve 100% fidelity. Time constraints at the end of the school year precluded collection of maintenance data, so intervention durability was not assessed.

Generalisation

Periodically, during the baseline and intervention phases, generalisation data were collected on participant behaviour during lunch in the cafeteria (shown as**Δ**in Figure 1). The cafeteria served as a generalisation setting as it was similar to, but slightly different from, the training setting. As on the playground, cafeteria observations lasted 10 minutes; also they followed similar data collection procedures. No intervention was provided in the cafeteria.

Interobserver agreement

For direct observation of the dependent variables, interobserver agreement (IOA) data were collected on of 35% of sessions across phases and participants.

Social vocalisation IOA

Interobserver agreement for the appropriate social vocalisation data was determined by comparing the time stamps from recordings of the behaviour. Whenever one data collector recorded the occurrence of a behaviour the other data collector had to record the behaviour within a five-second window to be considered an agreement. If the second data collector did not record the behaviour during that window, the incident was counted as a disagreement. Inter-observer agreement was calculated by dividing agreements by the total of agreements plus disagreements and multiplying by 100. Inter-observer agreement was 91% for appropriate social vocalisations (range 30% to 100%, *SD* = 16).

Peer response IOA

Interobserver agreement for positive, negative, and no peer response data were collected by identifying every instance of agreement on the occurrence of an appropriate social vocalisation and then comparing the positive, negative, and no peer response recordings. Any time both data collectors recorded the same peer response (that is, positive, negative,

or no) an agreement was counted. Any time data collectors recorded different responses or one data collector recorded a response that the other did not record, a disagreement was counted. IOA was calculated by dividing the number of peer response agreements by the total number of agreements plus disagreements, multiplied by 100. The mean IOA for peer responses was 74% (range 0% to 100%, *SD* = 36).

Implementation fidelity

Implementation fidelity data were collected on 46% of treatment sessions across the participants and interventionists. The average implementation fidelity was 97% (range 80% to 100%).

Social validity

To assess the social validity of the intervention, participants and interventionists were asked to provide feedback at the conclusion of the study. The participants completed the Children's Usage Rating Profile-Actual (CURP; Briesch & Chafouleas, 2009a) to determine how acceptable they found the procedures and how much they enjoyed the intervention. The CURP-Actual is scored on a 4-point Likert-type scale with higher scores indicating greater acceptability (Briesch & Chafouleas, 2009b).

The interventionists completed the Usage Rating Profile–Intervention (URP-I; (Chafouleas, Briesch, & Riley-Tillman, 2011) to indicate the ease of implementation of the intervention, the extent to which they found the procedures acceptable, and the extent to which they perceived a meaningful change in the behaviour of the participants. The URP-I is scored on a 6-point Likert-type scale with a higher score representing greater acceptance (Chafouleas, Riley-Tillman, Briesch, & Chanese, 2008; Briesch, Chafouleas, Neugebauer, & Riley-Tillman, 2013). To ensure the feasibility of this intervention, home-school collaboration and system support were reverse scored because they represented the need for outside support; this was consistent with the URP-I developer's recommendations.

Results

The purpose of this study was to evaluate the effects of BST social skills training on the social behaviour of students with ASD and the responses of their peers. Each of the research questions is addressed in detail, including a description of the visual analysis of the graphical data in accordance with the recommendations of Kratochwill and Levin (2014). Other quantitative (for example, Tau *U* effect size estimations) and qualitative data are presented to address relevant questions.

Frequency of social vocalisations

The frequency of social vocalisations by the three participants is presented in Figure 1. On average, the participants performed 7.81 (range 0 to 25) appropriate social vocalisations in a 10-minute observation during baseline, compared to 22.30 (range 8 to 39) for the composite of typically developing peers in the same period. During intervention phases, the participants' average increased to 21.88 (range 2 to 36) for an average change of

14.07 per 10-minute observation, an increase of 1.41 vocalisations per minute, compared to 19.32 (range 11 to 29) for the peer composite in the same period. The percentage of sessions in which participants' vocalisations fell within or above the 95% confidence interval of the peer composite during baseline was 12.5%. This number increased to 78% during intervention, indicating that in approximately three-fourths of sessions participants were interacting at roughly the same frequency as their typically developing peers. This change represents an average increase of 65.5% and a Tau *U* effect size estimate of 0.91, suggesting a large effect (Parker, Vannest, Davis, & Sauber, 2011).

Juan

The first panel in Figure 1 shows the frequency of Juan's social vocalisations in baseline and treatment phases. During baseline Juan's social behaviour showed a stable low trajectory with an average of 0.2 (range 0 to 1) social vocalisations per observation. On two generalisation probes in the cafeteria, Juan engaged in zero and two appropriate social vocalisations respectively. During recess Juan was observed wandering around the playground, swinging, or sitting on a bench alone as well as playing with other students from his special education resource class.

In the BST social skills training phase Juan's social vocalisations increased and became more variable, with an average of 20.24 (range 2 to 35) for a change of 20.22. The Tau *U* effect size estimate for this change was 0.97, suggesting a large effect. In 54% of treatment sessions, the frequency of Juan's social vocalisations reached or exceeded that of the 95% confidence interval for the peer composite (19.38 to 25.56). In the generalisation setting Juan's social behaviour remained low with the exception of one outlier (that is, session 15 with 12 appropriate social vocalisations), indicating a lack of generalisation.

Doug

In the second panel of Figure 1 Doug's social vocalisations in baseline and treatment are represented. In the baseline condition Doug's behaviour was highly variable and on two occasions reached the level of the 95% confidence interval for the peer composite (21.35 to 26.19). The average number of appropriate social vocalisations for Doug in baseline was 7.31 (range 0 to 25), with relatively low frequency in the generalisation setting. During baseline observations Doug was observed standing on the periphery of ongoing activities waiting to secure a ball or other piece of equipment, sitting reading books, guarding the ball containers from potential disorganisers, or securing his place as first in line long before the bell rang to signal the end of recess. On a few occasions Doug was invited to play games with, or referee games for, his typically developing peers. On these occasions he emitted many more appropriate social vocalisations than usual.

During the intervention phase Doug's appropriate social vocalisations increased to an average of 22.56 (range 14 to 36), up by 15.25, and remained variable. The Tau *U* effect size estimate for this change was 0.85, suggesting a large effect. For Doug, 40% of sessions in the social skills training condition reached or exceeded the frequency of 95% confidence interval of the peer composite. During this phase Doug frequently played basketball with other students rather than simply watching them. The interventionist noted that he was often very enthusiastic about the assignment to practice his new social skill with his peers.

Generalisation probes showed that Doug's social behaviour remained low in the cafeteria throughout the BST social skills training phase, indicating a lack of generalisation.

Elli

The frequency of Elli's social vocalisations during the baseline and treatment phases is charted on the third panel on Figure 1. During baseline Elli's appropriate social vocalisation rate was variable but relatively low, averaging 10.05 (range 0 to 21) per observation with 18% of the sessions reaching the level of the 95% confidence interval for the peer composite (17.29 to 25.10). In the generalisation setting Elli's appropriate social vocalisations remained low and stable. During recess Elli was observed wandering around the playground, talking with adults, or occasionally joining a large group game (for example, tag).

When BST social skills training was introduced, Elli's appropriate social vocalisations increased markedly and remained high though variable throughout the intervention. Her appropriate social vocalisations averaged 24 (range 19 to 32) in the intervention phase, indicating an increase of 14.79 and a Tau *U* effect size estimate of 0.91. In 100% of the intervention sessions, the frequency of Elli's appropriate social vocalisations reached or exceeded that of the peer composite. As with the other participants, Elli's social behaviour showed no meaningful improvement in the generalisation setting.

Generalisation

Figure 1 shows the frequency of appropriate social vocalisations in the cafeteria during lunch, the generalisation setting. None of the participants' appropriate social vocalisations in the cafeteria increased appreciably as a result of the intervention. Social behaviour of all of them was low during baseline and remained low during intervention, suggesting that BST social skills training had no effect on the dependent variables in contexts other than the playground.

Positive peer responding

Table 2 displays the percentage of appropriate social vocalisations that resulted in positive, negative, and no peer response. Both Juan and Doug experienced a decrease in the percentage of vocalisations that resulted in positive peer responses. Elli experienced a small increase of four percentage points. All participants experience a decrease in the percentage of vocalisations that resulted in negative peer responses; however these changes were very small (between 2 and 3.4 percentage points). Finally, Juan and Doug experienced a substantial increase in vocalisations with no response from peers (10 and 18 percentage points respectively), while Elli experienced a substantial decrease in vocalisations with no response.

Social validity

The average CURP rating for this intervention was 3.41 (range 3.21 to 3.67) out of a possible 4, representing an 85% acceptability rating of the BST social skills training intervention. The highest ranked dimension was Personal Desirability with an average score of 3.78 (95%, range 3.67 to 3.83). The next highest ranked dimension was Understanding with an average

Table 2. Percentage of vocalisations resulting in positive, negative, and no peer response.

	Positive Response			Negative Response			No Response		
Participant	BL	Tx	Change	BL	Tx	Change	BL	Tx	Change
Juan	62%	35%	−27	4%	2%	−2	51%	61%	+10
Doug	28%	15%	−13	4%	0.6%	−3.4	66%	84%	+18
Elli	51%	55%	+4	3%	0%	−3	50%	29%	−21

BL = baseline, Tx = treatment (i.e., BST social skills training)

score of 3.67 (94%, all participants scoring 3.67), and the lowest ranked dimension was Feasibility with a score of 2.80 (70%, range 2.29 to 3.5).

The average score on Acceptability was 5.17 (96%, range 4.78 to 5.56), for Understanding was 5.00 (83%, range 4 to 6), for Home-School collaboration was 4.50 (75%, range 3.67 to 5.33), for Feasibility was 5.00 (83%, range 4.67 to 5.33), for System Climate was 5.60 (93%, range 5 to 6), and for System Support was 2.67 (45%, range 2.33 to 3). The two dimensions that were reverse scored (Home School Collaboration and System Support) were the lowest rated dimensions, and Acceptability and System Climate were the highest rated. The average rating across all dimensions was 4.66 (78%). Without the two reverse scored dimensions, the acceptability rating became 5.19 (87%).

Discussion

The purpose of this study was to evaluate the effects of a BST social skills training approach on social vocalisations (topographical outcome) and peer responses (functional outcome) to identify possible discrepancies between topographical and functional outcomes among students with ASD in inclusive settings.

The research found that the BST social skills training approach improved the behaviour of all participants in using appropriate social vocalisations during lunch recess. Their social behaviour demonstrated marked improvements in level or trend, or both, during treatment. Although there was an obvious effect on the frequency of appropriate social vocalisations, there was also a mixed and sometimes counter therapeutic effect on peer responses, which may be a more important outcome in terms of the participants' quality of life. These results raise a question of whether participation in BST social skills training actually improved the participants' social skills, as defined by Elliott et al. (1995), or merely increased their social behaviour.

Dependent variables

Social interactions

Visual and statistical analysis of the data indicated that all participants' behaviour improved markedly with intervention. Across the three participants there were three demonstrations of an effect and no non effects, suggesting a functional relationship between the implementation of BST social skills training and the increase in appropriate social vocalisations. This finding was consistent with that of Ross and Sabey (2015) in which four participants demonstrated marked improvements in prosocial behaviour.

Peer responses

Each participant experienced some measure of improvement on peer responses. For all three the percentage negative responses decreased, and for Elli the percentage of positive responses increased and the percentage of receiving no response decreased. Consistent with Elliott et al. (1995) definition of social skills (that is, "Socially acceptable learned behaviours that enable a person to interact with others in ways that elicit positive responses and assist in avoiding negative responses") these changes may represent an improvement in the social lives of the participants. However, two participants also experienced some measure of decline on peer responses: Juan's and Doug's behaviour produced a lower percentage of positive peer responses and an increased percentage of no peer response. Assuming that positive peer attention functions as a reinforcer and negative peer attention functions as a punisher, all participants experienced less punishment, but Juan and Doug also received less reinforcement, which may possibly translate into some functional improvement in the quality of their social lives depending on which variable (that is, positive or negative peer responses) had a greater impact on them.

Increases in the percentage of vocalisations receiving no response from peers are difficult to interpret. By interacting more, participants were likely to experience more instances of no response from peers than other responses, as there had been more opportunities to interact. However, since increases in vocalisations receiving no response from peers were not related to decreases in vocalisations, the lack of peer response did not appear to have an extinguishing effect on social vocalisations.

Social interactions vs peer responses

When the dependent variable is defined as frequency of interactions (a measure easy to obtain, but probably less predictive of a good social life), BST social skills training clearly produced a meaningful effect, as participants' social vocalisations were similarly frequent to those of typically developing peers. However, when the dependent variable is designated as peer response (a measure more difficult to obtain and interpret but probably more predicative of a satisfying social life), whether the effect was meaningful is less evident. This discrepancy highlights the need for researchers to identify dependent variables that are predictive of the long-range outcome that is the goal of the intervention (that is, an improved social life).

Social validity

Participants endorsed BST social skills training as a socially valid intervention. The most highly endorsed dimension was Personal Desirability, indicating that participants liked participating in the intervention. CURP results suggested that participants' experience participating in BST social skills training was positive and that this intervention may be socially valid for primary school students with ASD.

Interventionists moderately endorsed the BST social skills training approach. This finding may under represent their endorsement of the intervention because the two dimensions that were not highly endorsed were reverse scored; thus interventionists may have intended higher scores to indicate greater endorsement across dimensions. When the two lowest dimension were removed, interventionists highly endorsed BST social skills

training. In terms of the overall rating of BST social skills training, it may be considered a moderately socially valid intervention for school-based interventionists.

Limitations and future research

Limitations to this study should be considered, particularly those which can be addressed by further research. First, due to time constraints maintenance data could not be collected and intervention durability was not addressed. Future research should include a maintenance phase.

Another limitation of the current study was its narrow scope. The school playground was the only setting targeted for intervention, but the utility of the intervention may be broader (for example, classrooms, library). Researchers need to identify situations beyond the playground where students with ASD demonstrate social deficits and follow the BST social skills training strategy in those contexts. Implementing the intervention in other contexts may also address the issue of generalisation. Training additional exemplars may enable the skill to generalise to novel contexts.

A third limitation was the relatively low IOA scores for the peer response dependent variable. Consequently, only tentative conclusions can be drawn about the functional effect of BST social skills training on peer response. Future researchers may wish to address this limitation by refining the measurement of peer responses and improving the training of data collectors to enable them to more reliably identify and record the different types of peer responses. This may be accomplished by making the peer responses the primary dependent variable of a study so that data collectors focus more on the peers. Additionally, data collectors may be trained to a higher standard regarding peer responses.

The results raise additional important issues, such as whether or not students with ASD are reinforced by positive peer responses, or can be conditioned, or taught, to be reinforced so that artificial contingencies can be removed. Additional research is needed to examine the functional role of peers in social interactions.

Implications for practice

The number of students diagnosed with ASD is increasing throughout the world (www.who.int). To fully benefit from their education, these students need effective social support. More teachers and educational psychologists need efficient and effective strategies to improve students' social skills; a BST social skills training approach may be a useful option to meet this need.

Also, schools are increasing emphasis on frameworks of multi-tiered systems of support for efficiently and effectively addressing students' academic and social needs (Sugai & Horner, 2009; Harlacher, Sakelaris, & Kattelman, 2014). A variation of the BST social skills training used in this study has demonstrated positive effects when paired with the very popular Tier 2 intervention Check-in Check-out (Ross & Sabey, 2015). BST social skills training may be another component intervention that can be used within multi-tiered systems of support framework to effectively serve all students.

Educational psychologists (EPs) are often expected to serve as "experts" to help address a problem beyond the capacity of the classroom teacher or other school personnel. Therefore, EPs may need to develop, implement, and evaluate the impact of interventions

that are designed to address social deficits of students with ASD. The results of this study suggest two aspects EPs might consider. Firstly, in choosing social skills interventions for students with ASD, *quality of life* outcomes must be considered in addition to the simpler *frequency of behaviour* outcomes often reported in the literature. Consulting the student, teachers, parents, and any others who frequently interact with the student is critical to determine what outcomes would constitute an improved quality of life for this individual. Secondly, because quality of life outcomes have not been carefully measured in the past, EPs may need to be creative in developing simple, practical, and efficient measurement procedures, recognising that they will likely vary among students. Developing these procedures will require EPs and other team members to focus on the ultimate goal of social skills interventions: producing an improved social life for the child with ASD.

Summary and conclusion

Students with ASD need support with social skills to succeed in school, fully benefit from their education, and be prepared for transition to more advanced education and eventual adulthood. Teachers need interventions that they can implement with minimal resources that do not detract from other teaching responsibilities. Researchers need interventions that address the functional impact of social skills training in terms of increasing positive social interactions and decreasing negative interactions to functionally improve social lives of individuals with ASD. BST social skills training may provide a promising option to go beyond topographic success and address functional needs, more fully addressing the ultimate purpose of PBS: to improve the quality of life for individuals with disabilities. Future researchers must go beyond the findings of this study to address ways that BST social skills training must be modified and refined to effectively promote the generalisation of newly acquired social skills to additional environments.

Disclosure statement

No potential conflict of interest was reported by the authors.

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